1949

EDSAC was one of the first recognizably modern computers in terms of design, but it nearly filled a room and needed 60 in- (152 cm-) long tubes of mercury to help with memory storage.

ON MARCH 28, BRITISH ASTRONOMER FRED HOYLE coined the term "**Big Bang**" on BBC radio, while explaining the Steady State Theory—his view that the Universe had an infinite past and an infinite future. He disagreed with the theory that all matter was created "in one big bang at a particular time in the remote past." In doing so, he unwittingly gave the name to an idea that would later gain almost universal acceptance.

Dutch–American astronomer **Gerald Kuiper** had established that the atmosphere of Mars was made of carbon dioxide and Saturn's rings were made of ice. In May, he **discovered Nereid**—the outermost moon of Neptune. Kuiper later attributed Nereid's eccentric orbit to its origin in an hypothesized ring of icy bodies beyond Neptune. The existence of this ring—called the **Kuiper Belt**—was confirmed in 1992.

On May 6, **EDSAC** (Electronic Delay Storage Automatic Calculator), a new computer at Cambridge University, UK, ran its first program. EDSAC could work through around 700 operations per second. It became the **first computer to routinely help scientists** with **complex calculations**.

In the US, rocket scientists successfully launched the **first mammal into space** (see 1948). Albert II—a rhesus monkey—went beyond the Earth's atmosphere when his V2 rocket reached a height of 81 miles (130.6 km). Albert II survived the flight, but he died on return to Earth due to parachute failure.

Monkey astronaut
The V2 rocket carrying a rhesus monkey went beyond the Karman Line—the boundary between the Earth's atmosphere and outer space.

FRED HOYLE (1915–2001)

British astronomer Fred Hoyle was one of the great scientific thinkers of the 20th century, and stimulated widespread interest in cosmology with his support of the Steady State Theory. This idea has been supplanted by the Big Bang Theory (see pp.344–45). However, Hoyle's Stellar Nucleosynthesis Theory prevails.

3 THOUSAND THE **NUMBER** OF **VACUUM TUBES** USED IN **EDSAC**

1950

Deployment of the myxoma virus was the first biological control method used for a mammal pest, dramatically reducing the number of rabbits in Australia.

THE YEAR 1950 SAW RAPID ADVANCEMENT in nuclear technology. On January 31, US President **Harry Truman** announced—largely in response to the Soviet Union's detonation of an atomic bomb in August 1949—that he had **authorized** the **development of a hydrogen bomb**. Its design would form the basis for all future thermonuclear weapons. A month later, at the University of California, nuclear chemist **Stanley Thompson** (1912–76) and his team created **californium**—the 98th element of the periodic table. Despite its instability, californium is still the heaviest element that does not quickly decay—and unlike most other ultraheavy elements, it can be made in quantities visible to the naked eye.

Computing science was racing forward too. The US had **ENIAC** (Electronic Numerical Integrator And Computer), which became the **first computer** to be used in **predicting weather**. It started the first 24-hour weather forecast service on March 5. In October, British computer scientist **Alan Turing** proposed a **test for artificial intelligence**.

Rabbits had become a national problem for Australia. A century and a half before, European settlers had brought rabbits with them for food. But in a land without predators, the rabbit population exploded, wreaking havoc on crops and native wildlife. When shooting, poisoning, and containment with fences failed to control them, **Frank Fenner** (1914–2010), a microbiologist at the Australian National University, oversaw the release of **myxoma—**a deadly virus that caused **myxomatosis**. It **curbed** the **rabbit plague**, and although rabbits were not

> " A **COMPUTER** WOULD DESERVE TO BE **CALLED INTELLIGENT** IF IT COULD **DECEIVE A HUMAN** INTO **BELIEVING** THAT IT **WAS HUMAN.** "
>
> **Alan M. Turing, British computer scientist,** 1950

March 28 British astronomer **Fred Hoyle** coins the term "**Big Bang**" on BBC Radio

May 1 Dutch–American **astronomer Gerald Kuiper** discovers **Nereid**, moon of Neptune

May 6 EDSAC runs its first program

June 14 First **mammal**—a rhesus monkey—in space

September Australian psychiatrist **John Cade** publishes first observations on the use of **lithium salts** to **control psychosis**

British chemist **Dorothy Hodgkin** (then Crowfoot) and her team publish the **molecular structure** of **penicillin**

December American chemists **James Arnold** and **Willard Libby** publish a method for **radiocarbon dating archaeological and geological samples**

January Dutch astronomer **Jan Oort** publishes his proposal about a **cloud of comets** at the edge of the Solar System

January 31 US President Truman orders development of **hydrogen bomb**

February 9 American physicist **Stanley Thompson** and team make the 98th element—**californium**

March 5 First 24-hour **weather prediction** by the computer **ENIAC**

August American chemist **Melvin Calvin** and his team publish a description of how **carbon** is used in **photosynthesis**